CHAPTER 10

RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

CHAPTER ESSENCE AT A GLANCE:
Master discrete and continuous random variables, probability mass functions (PMF), probability density functions (PDF), expectation, and variance properties.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Definition of Random Variable:** A real-valued function defined on the sample space S of a random experiment, mapping outcomes to real numbers X: S -> R (discrete if countable, continuous if an interval).

• **Probability Mass Function (PMF):** For discrete random variable X, p(x) = P(X = x) satisfies two fundamental axioms: (1) p(x) >= 0 for all x; (2) Total sum over all x equals 1 [sum p(x) = 1].

• **Probability Density Function (PDF):** For continuous random variable X, f(x) satisfies f(x) >= 0 and total area under curve integral from -infinity to +infinity equals 1; P(a <= X <= b) is integral from a to b of f(x) dx.

• **Mathematical Expectation and Variance:** Expected value E(X) = sum x p(x); Variance Var(X) = $E(X^2)$ - $[E(X)]^2$; E(aX + b) = a E(X) + b; $Var(aX + b) = a^2$ Var(X).

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Mathematical Expectation} \iff E(X) = \sum x_i \, p(x_i) \quad \text{or} \quad \int_{-\infty}^{\infty} x \, f(x) \, dx$$

Probability-Weighted Population Mean Center

$$\text{Variance Formula} \iff \text{Var}(X) = E(X^2) - [E(X)]^2 \geq 0$$

Second Central Moment Measure of Stochastic Dispersion

$$E(aX + b) = aE(X) + b \quad \text{and} \quad \text{Var}(aX + b) = a^2 \, \text{Var}(X)$$

Transformation Invariants of Expectation and Variance

$$\text{Discrete PMF Conditions} \iff p(x_i) \geq 0 \quad \text{and} \quad \sum_i p(x_i) = 1$$

Non-Negativity and Unit Sum Normalization Axioms

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** For a continuous random variable, the probability of taking any exact single point is identically zero: P(X = c) = 0; probabilities exist only over intervals.

• **Alert:** Variance is always non-negative: Var(X) >= 0; Var(aX + b) = a^2 Var(X), noting that constant b is completely dropped and scale a is squared.

• **Alert:** In formula card 3, remember that expectation is linear while variance scales quadratically.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define a Random Variable. Distinguish between Discrete and Continuous Random Variables. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** A random variable X is a real-valued function whose domain is the sample space S of a random experiment. **[1 Mark]**

• **Distinction:** Discrete random variable takes countable isolated values (e.g. number of heads in 3 coin tosses); Continuous random variable takes any value in an interval (e.g. lifetime of an electronic bulb). **[1 Mark]**

2. A random variable X has the probability distribution: X: 0, 1, 2, 3 with P(X): k, 2k, 3k, 4k. Find: (a) The value of k, (b) P(X >= 2), (c) P(1 <= X <= 2). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **(a) Find k:** Sum of all probabilities must equal 1: k + 2k + 3k + 4k = 1. 10k = 1, so k = 1 divided by 10 = 0.1. **[1.5 Marks]**

• **(b) P(X >= 2):** P(X = 2) + P(X = 3) = 3k + 4k = 7k = 7 * 0.1 = 0.7. **[1.5 Marks]**

• **(c) P(1 <= X <= 2):** P(X = 1) + P(X = 2) = 2k + 3k = 5k = 5 * 0.1 = 0.5. **[1 Mark]**

3. Find the Mean E(X) and Variance Var(X) of the number of heads obtained in two tosses of a fair coin. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Probability Distribution:** Let X = number of heads. X can be 0, 1, 2. P(X=0) = 1 divided by 4, P(X=1) = 2 divided by 4 = 1 divided by 2, P(X=2) = 1 divided by 4. **[1 Mark]**

• **Mean E(X):** sum x p(x) = 0*(1 divided by 4) + 1*(2 divided by 4) + 2*(1 divided by 4) = 0 + 2 divided by 4 + 2 divided by 4 = 4 divided by 4 = 1 head. **[1.5 Marks]**

• **$E(X^2)$:** sum x^2 p(x) = 0^2*(1 divided by 4) + 1^2*(2 divided by 4) + 2^2*(1 divided by 4) = 0 + 2 divided by 4 + 4 divided by 4 = 6 divided by 4 = 1.5. **[1 Mark]**

• **Variance:** $Var(X) = E(X^2) - [E(X)]^2 = 1.5 - 1^2 = 1.5 - 1 = 0.5$. **[0.5 Mark]**

4. If E(X) = 5 and Var(X) = 4, evaluate: (a) E(2X + 3), (b) Var(2X + 3). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) Expectation:** E(2X + 3) = 2 * E(X) + 3 = 2 * 5 + 3 = 10 + 3 = 13. **[1 Mark]**

• **(b) Variance:** $Var(2X + 3) = 2^2 * Var(X) = 4 \times 4 = 16$. **[1 Mark]**

5. State the two conditions that a function f(x) must satisfy to be a valid Probability Density Function (PDF). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Condition 1:** Non-negativity: f(x) >= 0 for all real values of x. **[1 Mark]**

• **Condition 2:** Total area under the curve equals 1: the definite integral of f(x) dx from -infinity to +infinity equals 1. **[1 Mark]**

6. Why is the probability P(X = c) equal to zero for a continuous random variable? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Explanation:** Probability for a continuous variable is measured by the area under the density curve over an interval. Since a single point c has an interval width of zero (dx = 0), the area under the curve above that single point is identically zero. **[2 Marks]**